

Model Evaluation — ML-Based Traffic Speed Prediction

This document describes the evaluation of the trained models published in this repository (`paper_artifacts/trained_models_base3y/`) accompanying the paper “Cloud-Fog-Edge Deployment of ML-Based Traffic Speed Prediction: An Empirical Analysis of Performance, Cost, and Scalability”.

Models

One multilayer perceptron per sensor direction (37 sensor directions; layer sizes 96-256-256-96-256, ReLU with dropout, ~142k parameters, RMSprop optimiser, MSE loss, float32). Each model ships with its fitted input (x) and output (y) RobustScaler as .pkl files. Models were trained on the initial three-year window (1 December 2020 to 1 December 2023) of five-minute average-speed records from Greater Manchester drakewell sensors, using 15 engineered features (cyclic time encodings and event flags). These are the base models from which all warm-start re-training scenarios in the paper are initialised.

Evaluation protocol

Each sensor’s data is split chronologically 80/20; the final 20% of the training window serves as the held-out test set (approximately the last seven months of 2023 for the three-year window). Metrics: mean absolute error (MAE, mph), root mean squared error (RMSE, mph), and mean absolute percentage error (MAPE, %). For context, each row also reports two properties of the sensor’s full speed series: the coefficient of variation of speed (Speed CoV, dimensionless) and the congestion frequency (share of five-minute intervals below 50% of the sensor’s 85th-percentile free-flow speed). A complementary common-test-set evaluation (identical held-out periods for all training scenarios) is summarised in the last section.

Per-sensor results — 3-year training regime

All rows in this table come from the **3-year initial training regime** (trained on 1 December 2020 to 1 December 2023); the held-out test set is the final 20% of that window, i.e. approximately mid-2023 to December 2023 (exact boundary varies slightly per sensor with data completeness). These are the published models in `trained_models_base3y/`.

#	Sensor	MAE (mph)	RMSE (mph)	MAPE (%)	Speed CoV	Congestion (%)	Test samples
1	1020 nw	1.95	2.92	8.02	0.128	0.63	61,132
2	1020 se	1.67	2.42	6.48	0.139	0.41	60,958
3	1056 e	1.36	1.92	6.06	0.086	0.11	59,752
4	1056 w	1.75	2.47	7.57	0.099	0.17	60,544
5	1063 e	1.45	2.50	5.76	0.097	0.43	61,213
6	1063 w	1.59	2.47	5.50	0.110	0.52	61,253
7	1071 ne	2.09	3.65	10.48	0.128	1.72	62,524
8	1071 sw	1.31	1.93	4.31	0.082	0.17	62,668
9	1163 nw	2.37	3.36	9.55	0.162	1.31	62,835
10	1163 se	2.97	4.74	16.52	0.190	5.41	62,843
11	1165 se	2.07	3.04	8.16	0.130	1.93	61,874
12	1173 n	1.83	2.63	6.38	0.437	21.36	58,707
13	1273 n	1.90	2.99	7.41	0.123	0.83	61,176
14	1276 e	1.56	2.24	6.85	0.154	1.04	60,057
15	1276 w	3.52	5.50	27.63	0.271	12.92	59,959
16	1385 n	2.10	3.24	11.30	0.190	6.18	58,161
17	1404 e	2.69	4.38	11.19	0.201	6.28	61,257
18	1404 w	2.60	3.93	10.09	0.216	6.53	60,197
19	1407 n	1.90	3.20	10.03	0.129	1.15	62,645
20	1407 s	2.14	3.12	9.71	0.171	3.80	62,573
21	1413 ne	1.58	2.27	6.06	0.102	0.34	62,277
22	1413 sw	1.64	2.25	5.92	0.100	0.11	62,174

#	Sensor	MAE (mph)	RMSE (mph)	MAPE (%)	Speed CoV	Congestion (%)	Test samples
23	1414 ne	1.54	2.13	6.18	0.124	0.39	60,300
24	1414 sw	1.41	1.98	5.66	0.114	0.29	60,317
25	1417 s	1.47	2.08	7.07	0.111	0.20	62,237
26	1418 e	1.92	3.28	9.09	0.118	1.44	55,437
27	1418 w	1.74	3.06	7.90	0.093	0.57	55,694
28	1420 nw	2.34	3.59	10.31	0.158	3.37	62,082
29	1420 se	1.91	2.90	8.64	0.128	0.92	61,899
30	1425 e	1.80	2.48	3.36	0.066	0.14	62,208
31	1425 w	2.16	4.38	6.25	0.080	0.51	62,376
32	1426 e	2.26	3.25	4.32	0.075	0.17	61,908
33	1426 w	3.75	6.26	12.68	0.171	3.26	62,685
34	1428 e	3.88	7.14	12.68	0.164	5.04	61,259
35	1428 w	2.17	3.25	4.12	0.066	0.06	61,158
36	1429 e	2.02	2.81	7.04	0.108	0.07	61,680
37	1429 w	2.00	2.90	7.04	0.108	0.25	60,874

Aggregate: mean MAE 2.07 mph (median 1.92); mean MAPE 8.47% (median 7.41%). The highest-error sensor directions on this test period are 1276 west (MAPE 27.6%) and 1163 southeast (16.5%). Per-sensor error correlates strongly with the two series properties in the table: on this test period, Spearman $\rho = 0.78$ between MAPE and Speed CoV, and $\rho = 0.80$ between MAPE and congestion frequency ($\rho = 0.79$ and 0.82 respectively for the 4.2-year cold-start regime evaluated on its own 2024-2025 held-out period, so the relationship is stable across both training regimes and test periods). The relationship is a strong rank trend rather than a per-sensor rule, and the exceptions are informative. Sensor 1173 north has the highest Speed CoV (0.437) yet only mid-range MAPE (6.4%): its whole-series CoV is dominated by a historical level shift (yearly mean speed 8.4, 14.3, 47.4, 46.8 km/h for 2020-2023), and in its stable regime it is highly predictable (a time-of-week average over the last four months of training data already achieves 5.9% MAPE). Conversely, 1276 west has stable yearly levels and a strong weekly rhythm, but large day-to-day spread within the same time-of-week slot (a time-of-week average still leaves 22% MAPE) - variation that no calendar-feature model can capture, which is what its 27.6% error reflects. This motivates a sharper variability measure: the within-time-slot coefficient of variation (standard deviation of speed after removing each sensor’s time-of-week mean profile, divided by mean speed), computed on a stable year of data (2023). Against this measure the correlation rises to Spearman $\rho = 0.94$, and the exceptions dissolve: 1276 west ranks first on both within-slot CoV (0.197) and MAPE, while 1173 north falls to rank 21 of 37, matching its error rank. In short, prediction difficulty is driven almost entirely by day-to-day spread at the same time-of-week; whole-series CoV is an imperfect proxy because it also counts historical level shifts, which a year-aware model absorbs. Note that the error ranking is period-dependent: on the 2024-2025 common-test-set windows (next section), platform 1404 becomes the dominant outlier (MAPE up to 86.5%). The sensor data shows why: both directions slowed progressively from ~ 57 km/h (August 2024) to ~ 33 km/h (January 2025) with variance roughly tripling - a months-long, bidirectional degradation consistent with long-term roadworks at the site - so models trained before the change are evaluated against a changed road.

Common-test-set scenario comparison

All training scenarios of the paper (3-year base, 4.2-year cold start, and progressive warm-start re-training windows PW3/PW6/PW12) were additionally evaluated on identical held-out periods. Mean MAE (mph) per evaluation window:

Scenario	Dec 2024 - Feb 2025	Oct 2024 - Feb 2025	May 2024 - Feb 2025
base (3y)	3.09	2.83	2.44
cold start (4.2y)	3.02	2.71	2.32
re-train PW3	2.91	2.68	2.35
re-train PW6	2.89	2.64	2.31
re-train PW12	2.90	2.61	excluded*

*PW12 trains up to September 2024, overlapping the nine-month window.

Warm-start re-training improves on both cold-start baselines in every window; prediction accuracy is invariant to the training platform (across repeated runs and platforms, the standard deviation of aggregate MAPE is below 0.3 percentage points in over 90% of measured cells).